The Macroeconomic Effects of Migration Structures

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1 Graphs

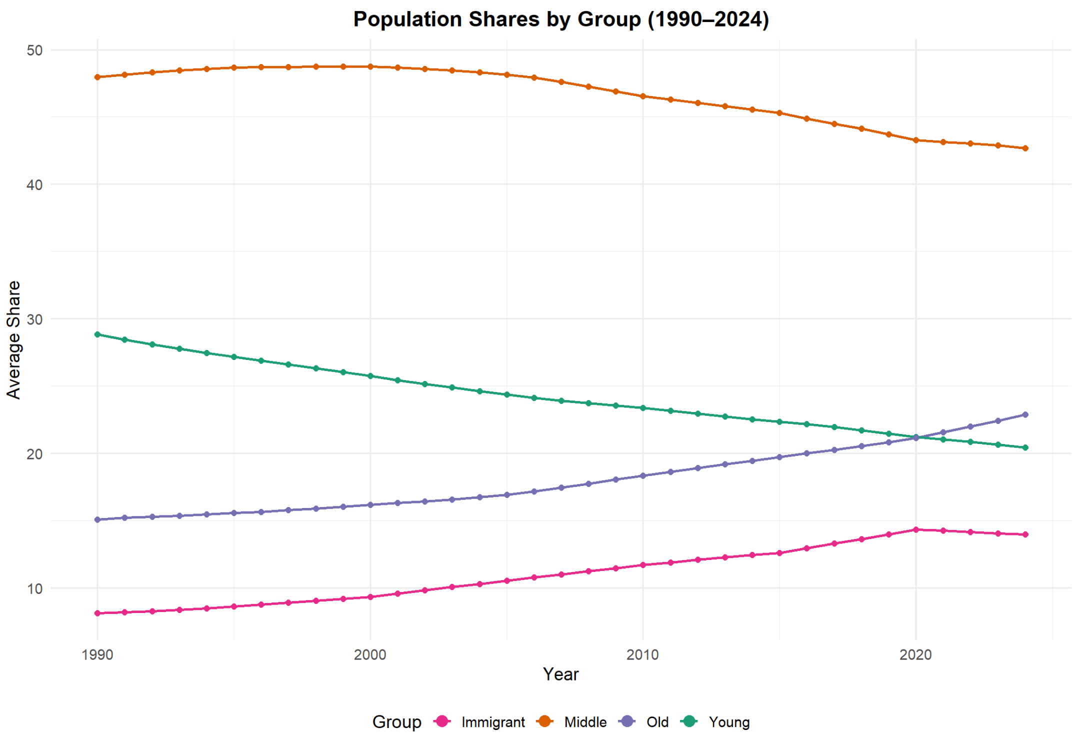


Fig. 1

Table 1: Effects of Population Variables on Macro Variables

Variable	young	mid	old	pop_g	immigrant
cpi	0.634 [0.334, 0.928]	-0.257 [-0.569, 0.053]	-0.027 [-0.309, 0.265]	0.340 [-0.877, 1.634]	-0.349 [-0.587, -0.115]
gdp_p	0.033 [-0.006, 0.071]	0.083 [0.039, 0.127]	-0.135 [-0.175, -0.097]	0.033 [-0.183, 0.268]	0.019 [-0.010, 0.049]
interestrate	0.686 [0.474, 0.901]	-0.180 [-0.396, 0.030]	-0.137 [-0.327, 0.059]	0.011 [-0.689, 0.708]	-0.369 [-0.543, -0.202]
inv	-0.271 [-0.468, -0.087]	0.398 [0.191, 0.611]	-0.306 [-0.490, -0.123]	0.037 [-0.846, 0.881]	0.179 [0.039, 0.332]
savingrate	-0.479 [-0.882, -0.117]	0.390 [-0.012, 0.827]	-0.202 [-0.586, 0.177]	-0.615 [-2.801, 1.172]	0.291 [0.014, 0.603]
unem	0.122 [-0.110, 0.376]	0.011 [-0.268, 0.278]	-0.013 [-0.258, 0.234]	0.535 [-0.873, 2.192]	-0.120 [-0.319, 0.063]

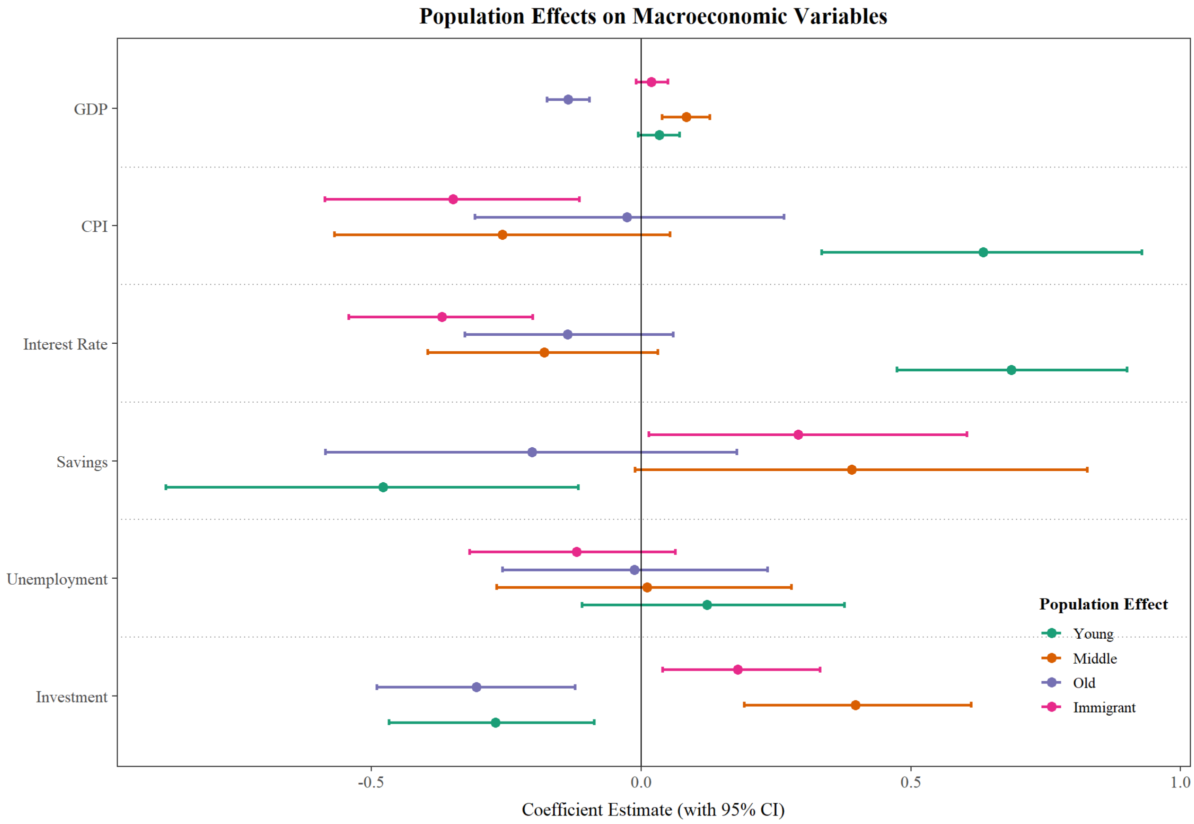

Population Effects on Macroeconomic Variables
GDP
CPI
Interest Rate
Savings
Unemployment
Investment
-0.5
0.0
0.5
1.0
Coefficient Estimate (with 95% CI)
Population Effect
Young
Middle
Old
Immigrant

Fig. 2

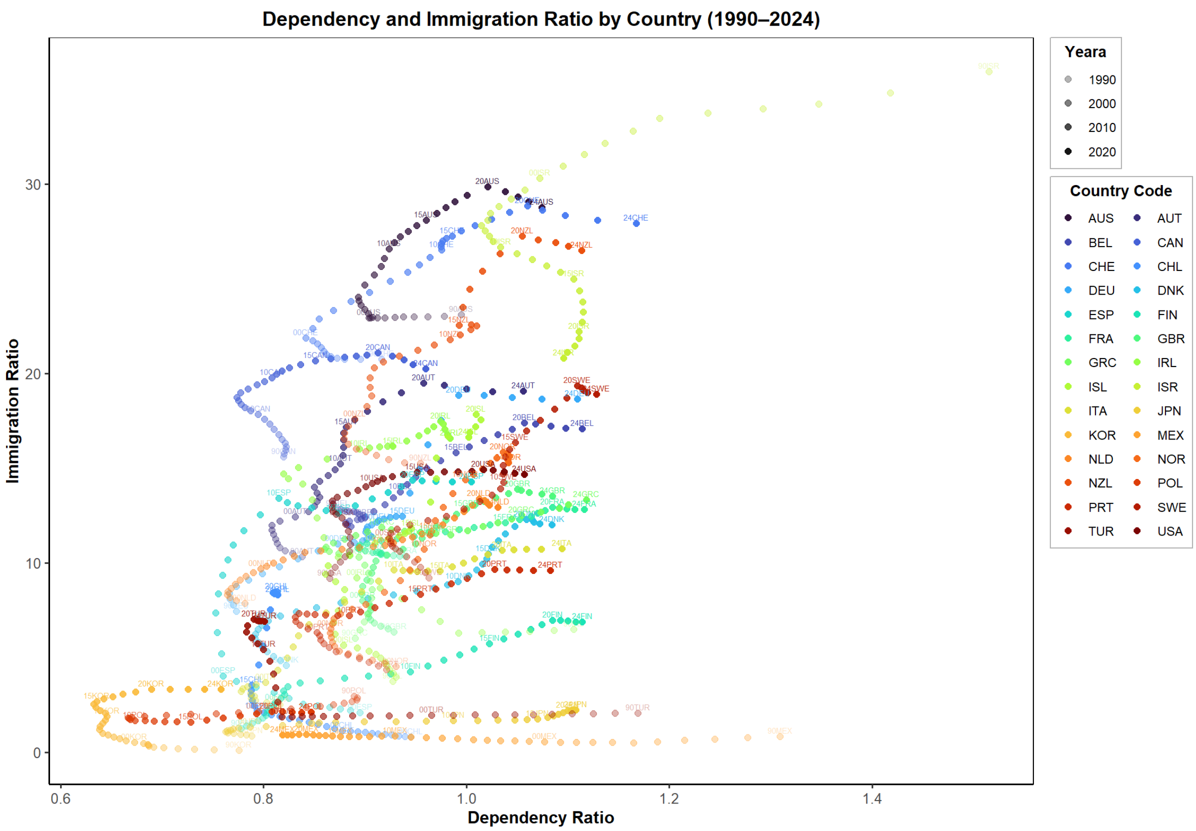

Dependency and Immigration Ratio by Country (1990–2024)
Immigration Ratio
Dependency Ratio
0
10
20
30
0.6
0.8
1.0
1.2
1.4
Yeara
1990
2000
2010
2020
Country Code
AUS
AUT
BEL
CAN
CHE
CHL
DEU
DNK
ESP
FIN
FRA
GBR
GRC
IRL
ISL
ISR
ITA
JPN
KOR
MEX
NLD
NOR
NZL
POL
PRT
SWE
TUR
USA

Fig. 3

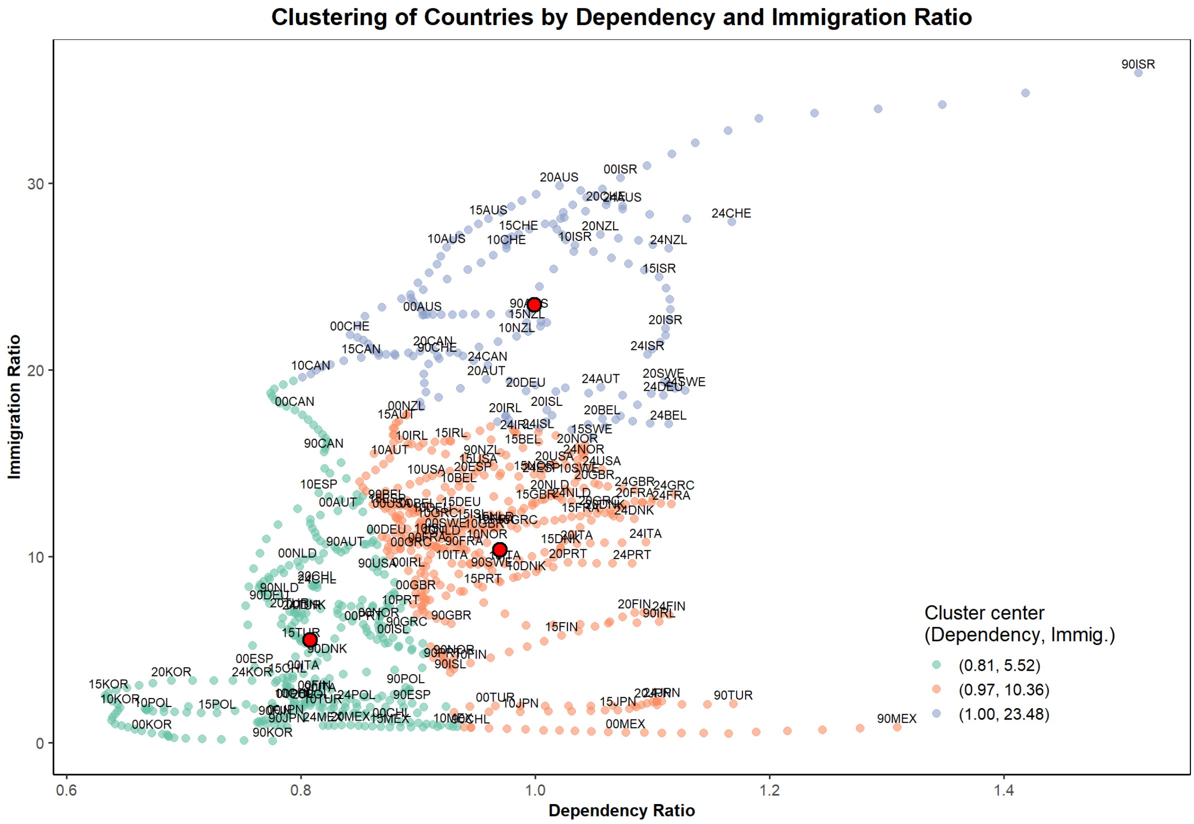
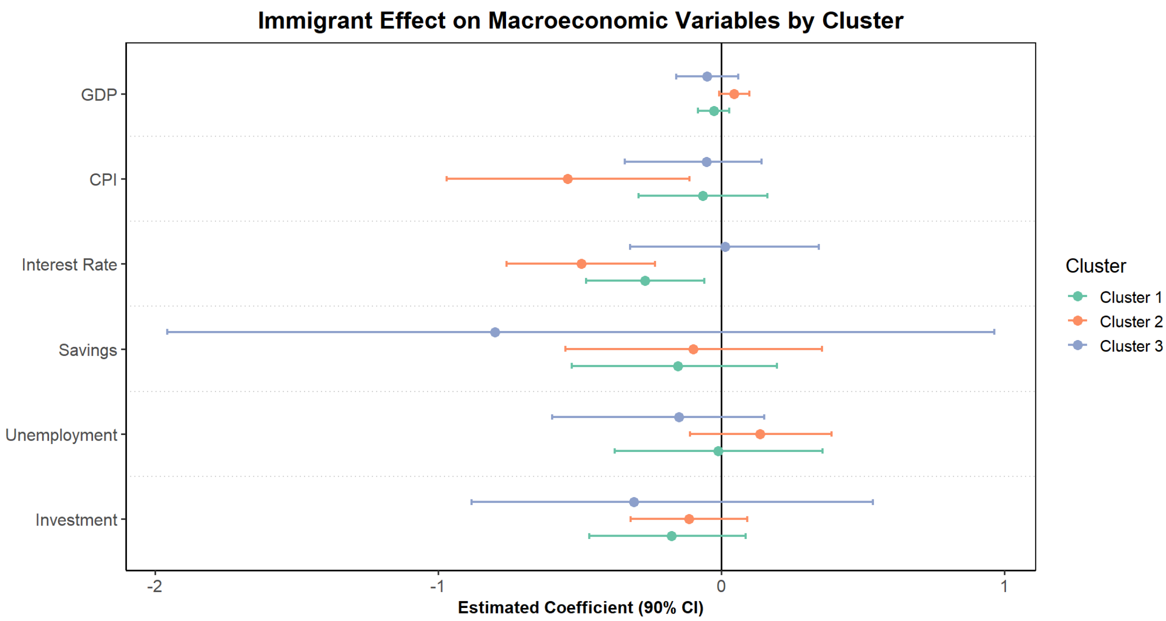


Fig. 4



Cluster	Dependency	Immigration
Cluster 1	Low	Low
Cluster 2	High	Low
Cluster 3	High	High

Fig. 5

2 Research Question

How do immigrants affect economic performance, and what factors drive the variation in their impact?

3 Hypotheses

H1. Countries with greater *immigration urgency* will experience a positive long-run effect of immigration on the economy

H2. Countries with greater *Immigrant inclusiveness* will experience a positive long-run effect of immigration on the economy(to be done)

H3. Diversity in immigrants' countries of origin will make a positive long-run effect of immigration on the economy(to be done)

4 Model Specification

4.1 Variables

We begin by defining the PVARX model, treating population structure as an exogenous variable, following Aksoy et al., 2019 with bayesian inference:

$$Y_{it} = (g_{it}, \pi_{it}, i_{it}, u_{it}, s_{it}, iv_{it})',$$

comprises six macroeconomic variables variables. The macro variables are real GDP growth g_{it}, the short-term interest rate i_{it}, CPI π_{it}, the investment-to-GDP ratio i_{it}, the personal savings-to-GDP ratio s_{it} and unemployment rate u_{it}.

The exogenous control vector is

$$B_{it} = \big(na_{1,it},\ na_{2,it}, na_{3,it}, pop\_gr_{it}\big)',$$

We divide the native population into three age-groups using cut-offs at ages 19 and 59: $Na_{1,it}$, $Na_{2,it}$, and $Na_{3,it}$ denote the shares of the native population aged below 19, between 20 and 59, and 60 or above, respectively. Then we define the native population shares

$$\{Na_{1,it},\ Na_{2,it},\ Na_{3,it}\}$$

are transformed to :

$$na_{1,it} = Na_{1,it} - imm_{it}, \tag{1}$$
$$na_{2,it} = Na_{2,it} - imm_{it}. \tag{2}$$
$$na_{2,it} = Na_{3,it} - imm_{it}. \tag{3}$$

This ensures no perfect collinearity.

4.2 Model Specification

To address overfitting in the moderately sized panels, we impose the following hierarchical prior structure in regression model:

$$Y_t = \alpha + A\,Y_{t-1} + B\,X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \varSigma), \tag{4}$$

where

- α is an 6×1 intercept vector.
- A is the 6×6 autoregressive coefficient matrix at lag 1.
- B is the 6×4 matrix of exogenous (population-effect) coefficients.
- $\varSigma$ is the 6×6 residual covariance matrix.

Minnesota–Gamma Prior

$$A_{ij} \sim \mathcal{N}\big(\mu_{ij},\, \sigma^2_{A,ij}\big), \qquad \sigma_{A,ij} = \begin{cases} \lambda_1, & i = j \quad \text{(own lag)}, \\ \lambda_2, & i \neq j \quad \text{(cross lag)}, \end{cases} \tag{5}$$
$$B_{k,i} \sim \mathcal{N}\big(0,\, \lambda_1\,\lambda_3\big), \tag{6}$$
$$\lambda_1, \lambda_2, \lambda_3 \sim \varGamma(0.1,\, 0.1). \tag{7}$$

where $\mu_{ij} = 1$ if $i = j$ (own-lag coefficient) and zero otherwise. The hyperparameters $(\lambda_1, \lambda_2, \lambda_3)$ control overall shrinkage, cross-equation shrinkage, and exogenous-variable shrinkage, respectively, and themselves are given weakly informative Gamma priors.

Error and Initial Conditions

$$\Sigma \sim \mathcal{IW}(\nu_0, \Psi_0),$$

with degrees of freedom $\nu_0 = M + 1$ and scale matrix $\Psi_0 = (0.02)^2 I_M$.

All posterior inference is carried out via Hamiltonian Monte Carlo using the `rstan` package, with 4 chains of 2000 iterations (1000 warmup) and an adapt constant of 0.9.

4.3 Estimated Shrinkage Hyperparameters

Table 2 reports the posterior means and 90% credible intervals for the three global shrinkage parameters. The estimated values are in line with those found in Bayesian VAR studies and confirm that our choice of Gamma(0.1, 0.1) priors produces sensible amounts of shrinkage.

Table 2: Posterior Estimates of Shrinkage Hyperparameters

Parameter	Posterior Mean	min	max
λ_1	0.4370	0.2451	0.8137
λ_2	0.1712	0.0783	0.3042
λ_3	0.1776	0.0770	0.3295

We use these values in our subgroup models as a fixed parameter.

4.4 Long-Run Effects

Once the posterior draws of the autoregressive matrix A and the exogenous-coefficient matrix B are obtained, we compute the long-run multiplier matrix Θ as

$$Y_t = \alpha + A\,Y_t + B\,X_t \quad \Longrightarrow \quad Y_t = (I - A)^{-1}\big(\alpha + B\,X_t\big) = (I - A)^{-1}\,\alpha \;+\; (I - A)^{-1}\,B\,X_t$$

$$\Theta = (I - A)^{-1}\,B$$

For the immigrant cohort, the long-run effect of a one-percentage-point increase in the immigrant(population cohort) share on the macroeconomic outcome. We derive these effects by computing for each posterior draw $(A^{(s)}, B^{(s)})$ the matrix

$$\Theta^{(s)} \;=\; (I - A^{(s)})^{-1}\,B^{(s)},$$

and then report the posterior mean and credible interval for each element Θ_{ij}.

Computation and Inference

- Draws $\{A^{(s)}, B^{(s)}\}_{s=1}^{S}$ are obtained via HMC in `rstan`.
- For each draw s, form $\Theta^{(s)} = (I - A^{(s)})^{-1}B^{(s)}$.
- Report $\mathbb{E}[\Theta_{ij} \mid \text{data}]$ and 90% credible interval $[\Theta_{ij}^{(0.025)},\, \Theta_{ij}^{(0.975)}]$.

Table 3: Data Source and Notes

Category	Variable	Source	Notes
Macroeconomic Indicators	GDP growth	IMF WEO,25	Domestic currency, constance price
	Real GDP growth	IMF WEO,25	—
	CPI	IMF WEO,25	—
	Short-term interest rate	FRED	3months or 90days interbank rate
	Investment-to-GDP ratio	IMF WEO,25	Private
	Savings-to-GDP ratio	IMF WEO,25	Private
	Unemployment rate	IMF WEO ,25	—
Demographic Indicators	Immigrant ratio	UNPD IMS,26	Five-year linear interpolated
	Native age structure	UNPD WPP,25	July 1
Immigrant Indicators	some indicators(not decided)	MIPEX	—
	immigrant unemp rate	ILO stat	—

References

Aksoy, Y., Basso, H. S., Smith, R. P., and Grasl, T. (2019). Demographic structure and macroeconomic trends. *American Economic Journal: Macroeconomics*, 11(1):193–222.